

pH Compliance.

What if pH deviation could be predicted before the batch is manufactured?

Random Forest · 300 Trees · Lot Release Prediction



80.4%

Recall

11

Process Features

84.1%

Accuracy

0.918

ROC-AUC

In oral liquid pharmaceuticals, pH is not a formulation preference — it is a release criterion. Outside spec means no release.



Regulatory Cost

Every pH deviation triggers a deviation investigation, documentation and potential regulatory scrutiny



Reprocess Risk

Failed lots require reblending, re-analysis, and batch schedule delay — if salvageable at all



Root Cause Gap

pH is the outcome of a system: acid, buffer, water, mixing — no single variable explains deviation

58.6%

of lots failed pH specification — requiring reprocess or disposal. The spec window is 4.5–5.5 pH. Final pH ranged from 1.89 to 6.73.

Old Thinking

pH deviation is found at release:

"The lot failed pH. Raise a deviation."

Manufacture the batch. Wait for analysis.
Fail QC. Investigate root cause.
Reprocess if feasible. Document everything.
Repeat next campaign.



New Thinking

Risk is scored before manufacture:

"How likely is this batch to comply?"

Input the planned recipe and conditions.
Score probability of pH compliance.
Adjust formulation before synthesis begins.
Prevent the deviation, not just document it.

The batch that fails pH always had a recipe that could have predicted it — if the right model was in place.

Eleven variables spanning formulation chemistry, purified water quality, and mixing process conditions:

FORMULATION	API %, Citric acid %, Citrate buffer %, Sweetener %, Preservative % <i>Master Batch Record — the recipe inputs that directly drive pH chemistry</i>
WATER	Purified water pH, Previous lot pH <i>QC lab records — water quality and manufacturing line carryover effect</i>
PROCESS	Mixing temperature (°C), Mixing time (min), Agitation speed (RPM) <i>Process historian / SCADA — physical mixing conditions per batch</i>
SEQUENCE	Addition order (Buffer→Acid→API / Acid→Buffer / API→Buffer) <i>Manufacturing SOP — ingredient sequence affects final pH significantly</i>

Addition order matters: Buffer→Acid→API (order 1) achieves 45.0% approval rate vs 34.6% for Acid→Buffer→API (order 2). Sequence is chemistry.

All 11 variables exist in systems you already have — no new sensors, no new instruments. Just connected data.

pH is not a linear function of any single ingredient — it's a system response. Random Forest captures interactions that no formula sheet can show.

No Scaling Required

Tree-based models use threshold splits — a % concentration and a temperature in Celsius compete equally. No StandardScaler, fewer deployment failure points.

Ensemble Variance Control

300 trees, each trained on a random data subset and random feature subset. Individual trees overfit; the majority vote does not. AUC jumps from 0.75 (single tree) to 0.92.

Non-linear Chemistry

High citric acid + high buffer = different outcome than high acid + low buffer. RF captures these multiplicative interactions without explicit feature engineering.

Dual Interpretability

Gini importance for fast screening; permutation importance for rigorous test-set validation. Two measures that converge on the same story give high confidence.

84.1%

Test Accuracy

0.918

ROC-AUC

80.8%

F1 (Approved)

80.4%

Recall (Approved)

Feature Importance — Gini & Permutation

Gini

Permutation

citric_acid_pct	23.7%	Δ -24.2%
citrate_buffer_pct	21.6%	Δ -17.7%
agitation_rpm	3.8%	Δ -1.7%
prev_lot_ph	9.2%	Δ -3.8%
mixing_time_min	7.5%	Δ -3.1%
water_ph / temp / rest	29.0%	Δ <1.7%

Confusion Matrix (555 test batches)

	Pred: Fail	Pred: Pass
Actual: Fail	282 (TN)	43 (FP)
Actual: Pass	45 (FN)	185 (TP)

45 false negatives — batches that would fail but the model predicted pass. Each one a potential deviation event.

Both Gini and Permutation importance agree: citric acid and citrate buffer dominate. Chemistry confirms the model.

Citric Acid 23.7% Gini

→ Primary lever. Tighten citric acid specification range in the Master Batch Record — small deviations in acid % produce large pH swings at this buffer level.

Citrate Buffer 21.6%

→ Counter-lever. Buffer concentration must be co-controlled with acid. A high-acid formula without adequate buffer will systematically fail — it is a ratio problem.

Previous Lot pH Δ -3.8%

→ Carryover effect is real. When the previous lot pH is at the lower boundary (≤ 4.6), the next batch carries elevated deviation risk — flag it in the scheduling system.

Agitation RPM Δ -1.7%

→ Low agitation (< 200 RPM) is associated with inadequate homogenization. Set a validated minimum — 450 RPM appears consistently in in-spec batches.

Addition Order 1 vs 2

→ Buffer→Acid→API (order 1) achieves 45.0% approval vs 34.6% for Acid→Buffer. Prefer order 1 for formulations near the acid/buffer boundary.

What-If Simulator — Pre-Batch Risk Scorecard

08

Same equipment. Same operator. Three recipe decisions. Scenario C reveals what process alone cannot fix.

Batch A — In-Spec	Batch B — Out-of-Spec	Batch C — Intervention
Citric acid0.8%	Citric acid1.7%	Citric acid1.7%
Buffer4.0%	Buffer1.2%	Buffer3.5% ↑
Agitation600 RPM	Agitation150 RPM	Agitation550 RPM ↑
Prev lot pH5.0	Prev lot pH4.5	Prev lot pH4.5
Temp25°C	Temp30°C	Temp30°C
OrderBuffer→Acid	OrderAcid→Buffer	OrderAcid→Buffer
 56.3% approval probability	 11.7% approval probability	 1.0% approval probability

Scenario C: increasing buffer and agitation makes things worse (11.7% → 1.0%). With citric acid at 1.7%, process corrections cannot override a formulation root cause. The acid must be reduced.

The simulator makes the chemistry visible before the batch. That's the difference between a prediction and a decision.

This project helps to understand how to predicting pH.

preventing the deviation before the batch is made.

300 trees, each seeing a different slice of the data,
vote on whether this recipe will comply. Chemistry meets ensemble learning.

Feel free to download the full project: dataset, notebook, and simulator.

Change the acid. Adjust the buffer. Challenge where the model draws the line.

Because pharmaceutical quality has no finish line.

